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EXPERIMENT – 2

CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB

PROGRAM:

```
clc;
clear;
close all;
Am=1;
fm=500;
Ac=5;
fc=5000;
fs=20*fc;
t=0:1/fs:4/fm;
m = Am*sin(2*pi*fm*t);
c = Ac*sin(2*pi*fc*t);
mu = Am/Ac;
am = Ac*(1+mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);
dem = abs(hilbert(am)); dem = dem-mean(dem);
N=length(am); f=linspace(-fs/2,fs/2,N);
AM_FFT=abs(fftshift(fft(am)))/N;
USB=fc+fm; LSB=fc-fm;
Pc=Ac^2/2; Psb=(mu^2*Pc)/4;
figure('Name','AM Modulation','NumberTitle','off');
subplot(3,2,1), plot(t,m), title('Message'), grid on
subplot(3,2,2), plot(t,c), title('Carrier'), grid on
subplot(3,2,3), plot(t,am), title('AM Signal'), grid on
```

subplot(3,2,4), plot(t,dem), title('Demodulated'), grid on

subplot(3,2,5), plot(f,AM_FFT), xlim([fc-3*fm fc+3*fm]), title('Sidebands'), grid on

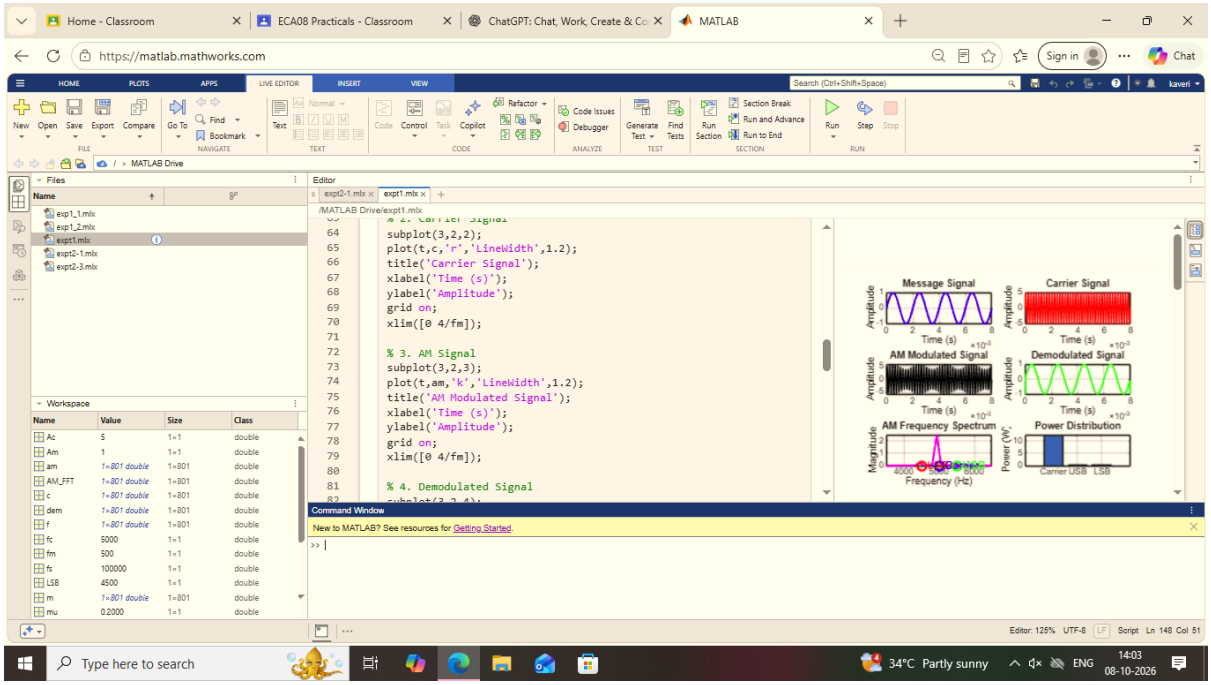
subplot(3,2,6), bar([Pc Psb Psb]), set(gca,'XTickLabel',{'Carrier','USB','LSB'}),

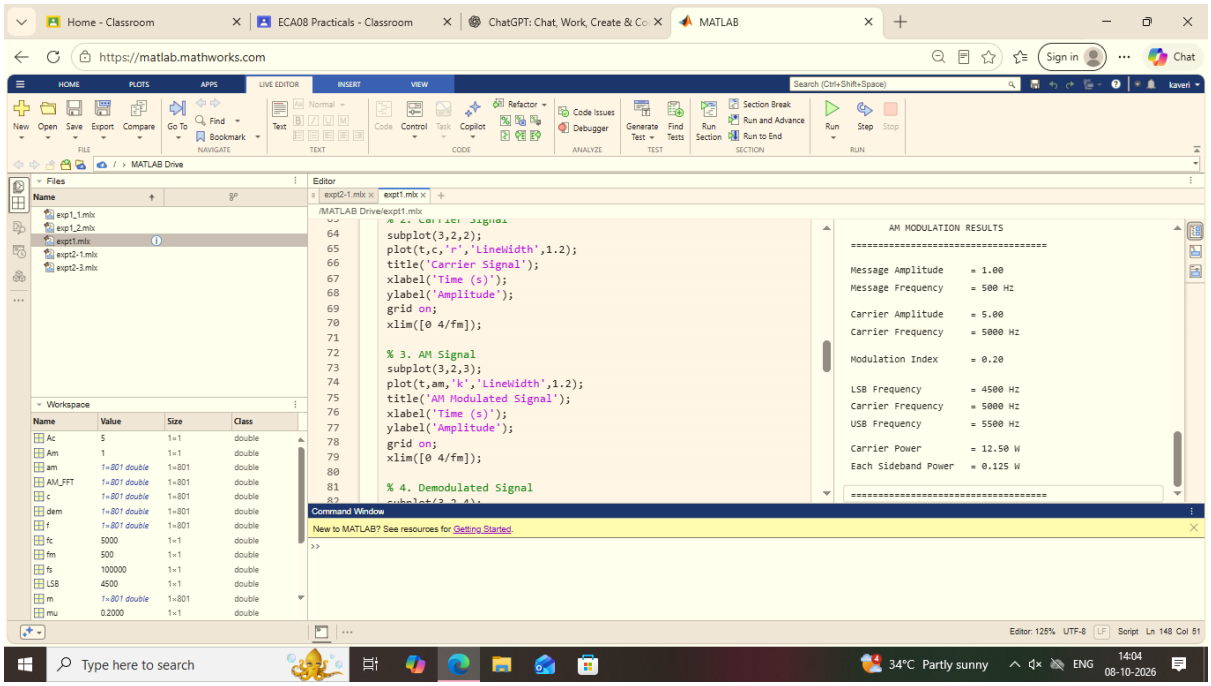
title('Power'), grid on

fprintf('μ=%.2f | USB=%d Hz | LSB=%d Hz | Pc=%.2f W | Psb=%.3f

W\n',mu,USB,LSB,Pc,Psb);

OUTPUT:




Home - Classroom
ECA08 Practicals - Classroom
ChatGPT: Chat, Work, Create & Co
MATLAB
https://matlab.mathworks.com
Sign in
Chat
HOME
PLOTS
APPS
LIVE EDITOR
INSERT
VIEW
MATLAB Drive
Files
Name
expt1_1.mlx
expt1_2.mlx
expt1.mlx
expt2-1.mlx
expt2-3.mlx
Workspace
Name
Value
Size
Class
Editor
expt2-1.mlx
expt1.mlx
/MATLAB Drive/expt1.mlx
subplot(3,2,2);
plot(t,c,'r','LineWidth',1.2);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
xlim([0 4/fm]);
% 3. AM Signal
subplot(3,2,3);
plot(t,am,'k','LineWidth',1.2);
title('AM Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
xlim([0 4/fm]);
% 4. Demodulated Signal
AM MODULATION RESULTS
Message Amplitude = 1.00
Message Frequency = 500 Hz
Carrier Amplitude = 5.00
Carrier Frequency = 5000 Hz
Modulation Index = 0.20
LSB Frequency = 4500 Hz
Carrier Frequency = 5000 Hz
USB Frequency = 5500 Hz
Carrier Power = 12.50 W
Each Sideband Power = 0.125 W
Command Window
New to MATLAB? See resources for Getting Started.
Type here to search
34°C Partly sunny
ENG
14:04
08-10-2026